List of Experiments (Image and Speech Processing)

1. Display following image operation in MATLAB/Python - i) Histogram image ii) Low pass filter mage iii) High pass image.

2. Write a MATLAB/Python program to read 'rice.tif' image, count number of rice and display area (also specific range), major axis length, and perimeter.

3. Write a MATLAB/Python program to read an image and perform convolution with 3X3 mask.

4. Write a MATLAB/Python program to read an image and perform Lapliciant filter mask.

5. Write a MATLAB/Python program to identify horizontal, vertical lines from an image. .

6. Write a MATLAB/Python program to Character Segment of an image.

7. For the given image perform edge detection using different operators and compare the results.

8. Write a MATLAB/Python program to read coins.png, leveling all coins and display area of all coins.

9. Display following image operation in MATLAB/Python - i) Threshold image ii) Power enhance contract image iii) High pass image.

10. Perform image enhancement, smoothing and sharpening, in spatial domain using different spatial filters and compare the performances

11. Perform image enhancement, smoothing and sharpening, in frequency domain using different filters and compare the performances

12. Write a MATLAB/Python program to separation of voiced/un-voiced/silence regions from a speech signal.

13. Write a MATLAB/Python program and plot multilevel speech resolution.

14. Write a MATLAB/Python program to recognize speech signal.

15. Write a MATLAB/Python program for text-to-speech conversion and record speech signal.